

JORGE MARTINEZ

Calle de Albalá 9, Madrid, Spain 28037

+34 601 368 929 [✉ jorgemartinezgil117@gmail.com](mailto:jorgemartinezgil117@gmail.com) [in LinkedIn](#) [🌐 Website](#) [🐙 Github](#)

Experience

Independent AI and Software Development Consultant

Jan 2025 – Actualidad

- Developing generative AI projects, including multi-agent RAG systems and image generation workflows using ComfyUI.
- Designing, fine-tuning, and deploying machine learning models such as Transformers, CNNs, and XGBoost.
- Implementing full-stack applications with Python, C#, and React, and orchestrating deployments on Azure.

Universidad Carlos III de Madrid

Feb 2024 – Jul 2024

Language Processors Teacher

Madrid, Spain

- Instructed over 80 students in practical compiler construction classes, focusing on transpiler from C to Lisp and from Lisp to Forth using Bison and C.
- Conducted hands-on coding sessions covering all essential steps of compiler construction.
- Responsible for correcting and grading 25% of each theoretical exam.
- Received a Teaching Excellence Acknowledgment with a 4.13/5 student evaluation score.
- My students achieved a 94% pass rate, contributing to a 76% overall pass rate.

ATRIA Innovation

Feb 2023 – Apr 2023

Software Engineer Intern

Zaragoza, Spain

- Contributed to the development of a Python program that employs Convolutional Neural Networks for the detection of manufacturing issues in car images.

Education

Universidad Carlos III de Madrid

Sep. 2023 – Current

Double Master Degree in Computer Science and Applied Artificial Intelligence

Madrid, Spain

Universidad de Zaragoza

Sep. 2019 – Jun 2023

Bachelor's degree in Computer Science

Zaragoza, Spain

George Mason University

Aug. 2022 – Dec 2022

Bachelor's degree in Computer Science

Fairfax, Virginia

Skills

- C++ | Go | Python | JavaScript | NodeJS | Kotlin | Algorithm Design and Analysis | Data Structures | Android Studio
- Git | GitHub | Docker | Kubernetes | Google Cloud | Distributed Systems | Database Management | SQL
- Linux | Computer Architecture | Machine Learning | AI | Spanish (Native) | English (Fluent) | German (Intermediate)

Projects

DOVS Model for Navigation in Dynamic Environments | *Python*

- Developed a Python-based parallelizable implementation that effectively models real-time dynamic 2D environments with multiple agents.

LOL E-Sports Bets | *Kotlin, Go, Android, Google Cloud*

- Developed an Android app in Kotlin, featuring a coin-based betting system tailored for e-Sports League of Legends enthusiasts.
- Implemented a Go backend server adept at handling bets in real-time, enabling seamless communication with the app.
- Deployed the app on Google Cloud using a Compute Engine VM for the backend and Cloud SQL for the database.

March Machine Learning Mania 2024 | *Python, Machine Learning*

- Developed a predictive model to accurately estimate the outcomes of NCAA basketball tournament games for the Kaggle competition "March Machine Learning Mania 2024."
- Conducted comparative analysis of machine learning techniques, including XGBoost and neural networks.

Activities & Leadership

Applied Artificial Intelligence Group, UC3M

Sep. 2023 – Present

Research Member

Madrid, Spain

- Contributed to a research project alongside faculty, leading to the publication of the article "Cloud-based System for Monitoring Event-Based Hydrological Processes Using a Dense Sensor Network and NB-IoT Connectivity."

ZAMUN (Žilina Model United Nations)

2019

Participant

Žilina, Slovakia

- Took part in the international debate program promoted by the UN, representing Brazil in the topic "Economic Instability and the Protection of the Environment" and "Asymmetric warfare on the Internet".

Basketball Enthusiast

Ongoing

Player

Zaragoza, Spain

- Plays basketball in a local league, fostering teamwork and competitive spirit.
- Served as high school basketball team captain for one year, developing leadership skills.